Data Analysis, Gradient Descent, and Stochastic Gradient Descent : A Case Study

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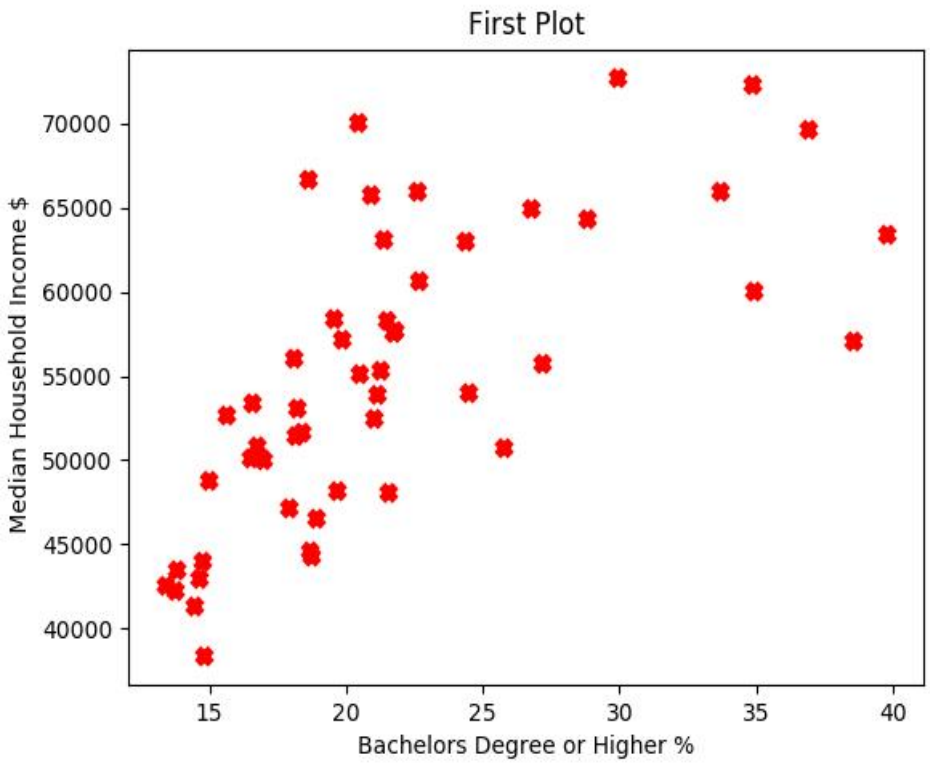
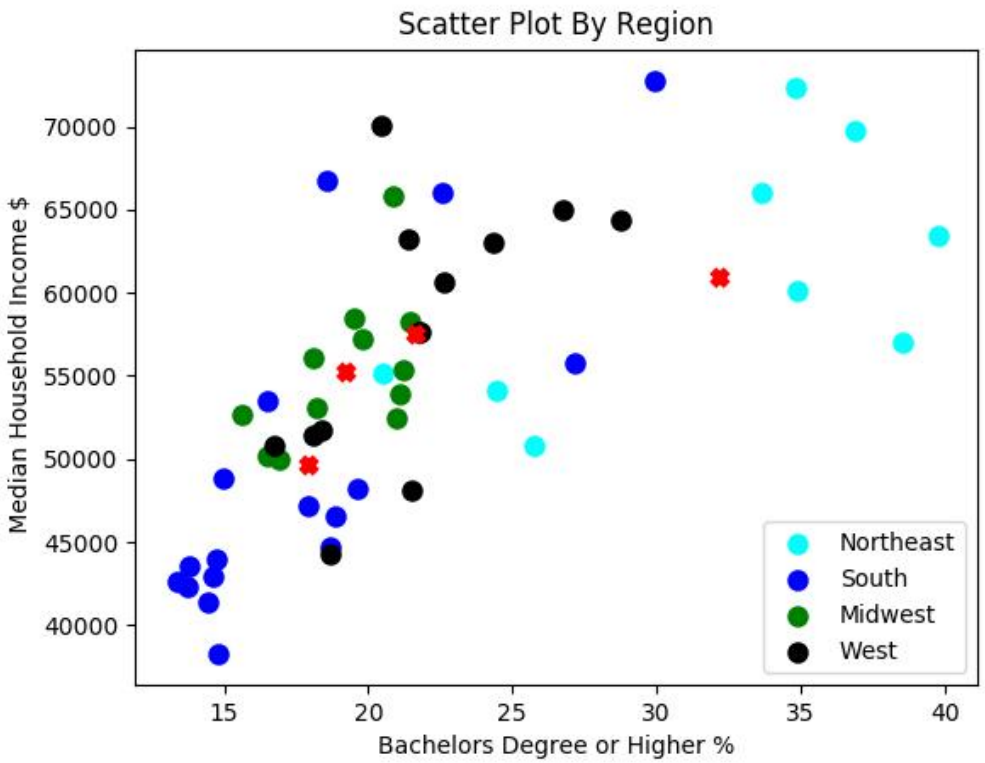
METHODS

- Collect data about 50 states + District of Columbia for Median Household Income, Median Educational Attainment (At least a Bachelor's Degree) between 2012 and 2016.
- Create Scatter plot of 50 states + Washington D.C by Median Household Income and Educational Attainment. Label the states by regions. REGION 0 : Northeast, REGION 1:South, REGION 2: Midwest, REGION 3: West.
- Create scatter plot that highlights regions and find the centroid of the clusters
- Standardize the data and classify the regions using K Nearest Neighbor
- Create a linear regression line using Gradient Descent and Stochastic Gradient Descent

SCATTER PLOTS





K- Nearest Neighbor Classification

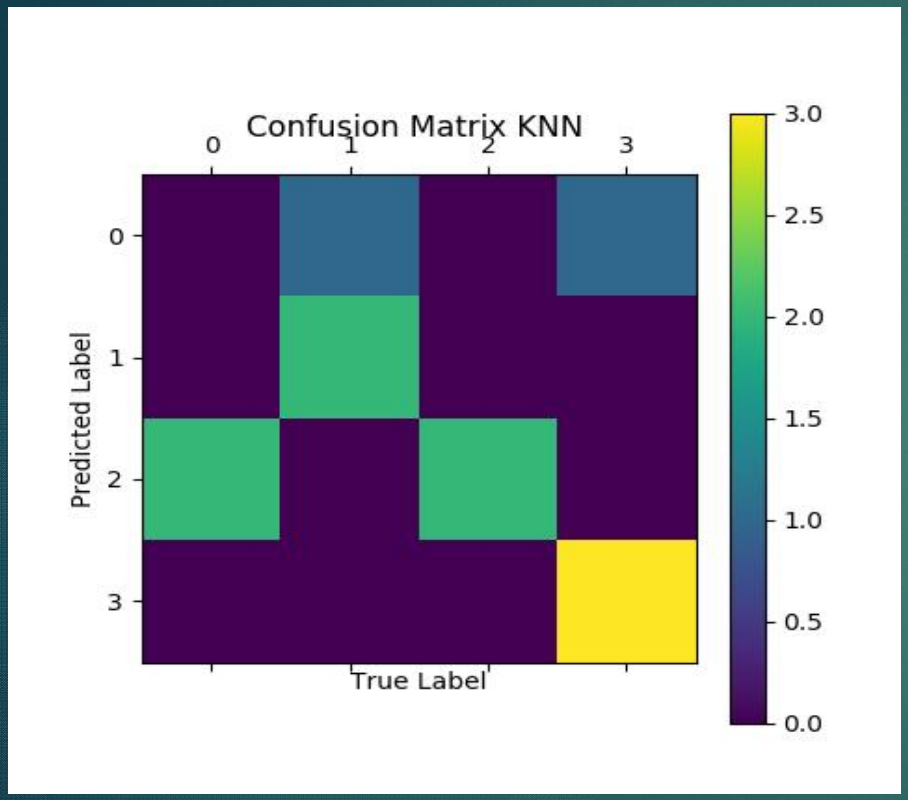
- X = EDUCATION, INCOME
- Y = REGION

- Supervised learning
- K Nearest neighbor uses the majority vote to assign a label
- Use some kind of distance metric to calculate the distance between two points. Most commonly, Euclidean distance:

$$d(x, x') = \sqrt{(x_1 - x'_1)^2 + (x_2 - x'_2)^2 + \ldots + (x_n - x'_n)^2}$$

- Take the 1-Nearest neighbor and assign this class label
- Test size is 20% of Data set

K-NN Confusion Matrix




```
[0, 1, 0, 1],
[0, 2, 0, 0],
[2, 0, 2, 0],
[0, 0, 0, 3]]
```

Y True = [3, 0, 0, 2, 2, 3, 3, 2, 1, 2, 1]
Prediction= [3, 3, 1, 2, 0, 3, 3, 2, 1, 0, 1]

64% Classification Accuracy

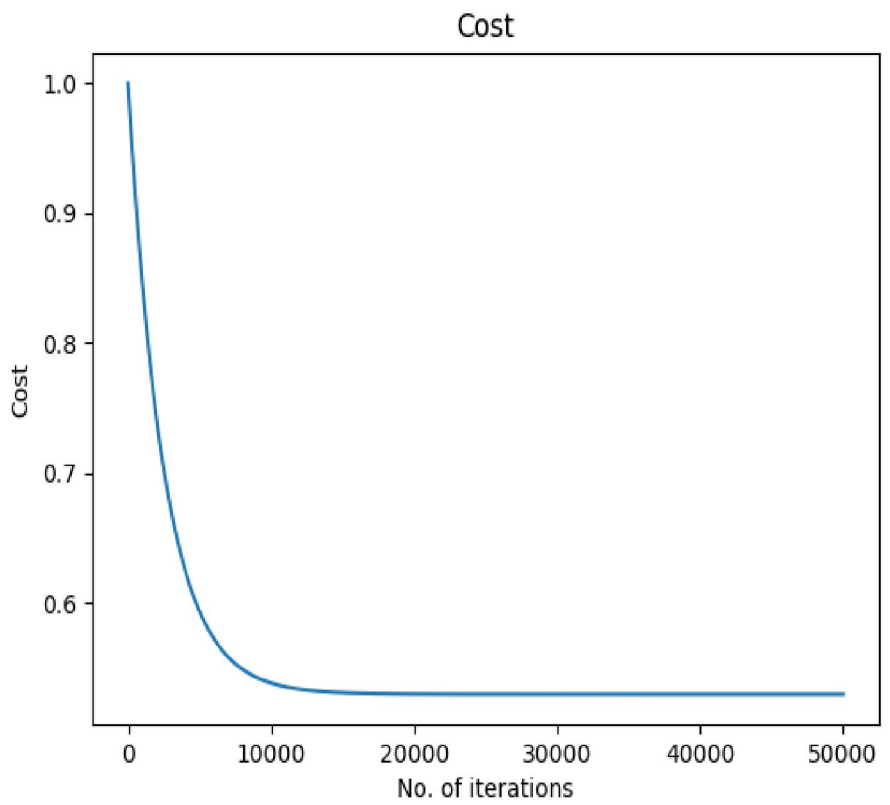
Linear Regression with Gradient Descent

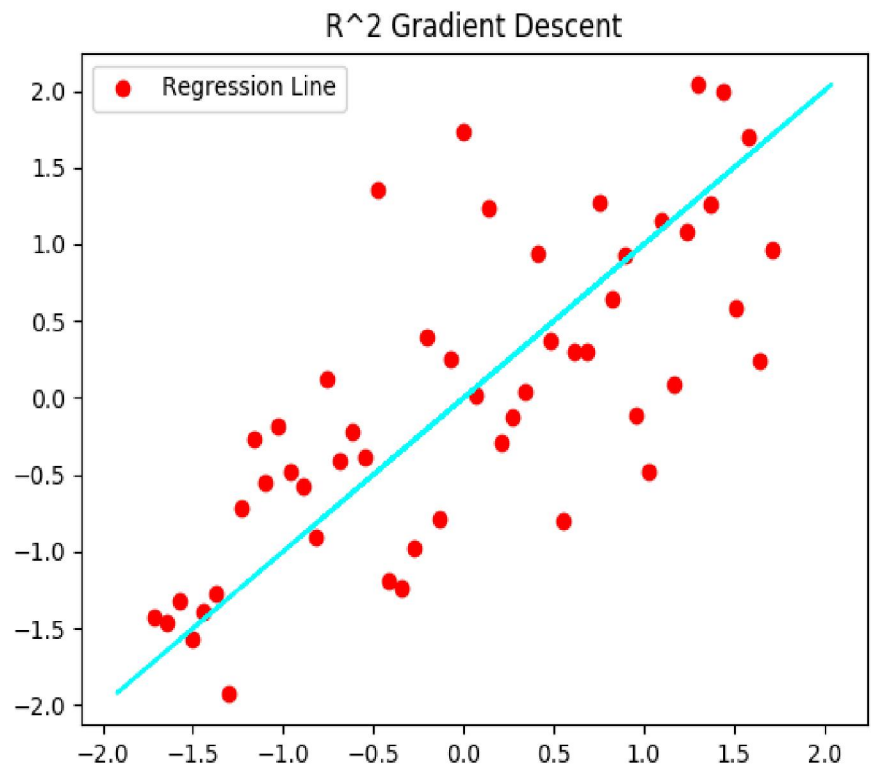
X = Education

Y = Income

- Optimization of parameters to get the best fit line. Keep a constant Learning Rate.
- Goal is to minimize the cost function J, or the Mean Squared Error
- $y = m\,x + b$
- $J = \sum_{i=1}^{N} \frac{(Y-y)^2}{N}$
- $\frac{\partial J}{\partial m} = \frac{-2}{N} \sum_{i=1}^{N} x(Y - y)$
- $\frac{\partial J}{\partial b} = \frac{-2}{N} \sum_{i=1}^{N} (Y - y)$
- $m = m - \alpha \frac{\partial J}{\partial m}$
- $b = b - \alpha \frac{\partial J}{\partial b}$

Gradient Descent





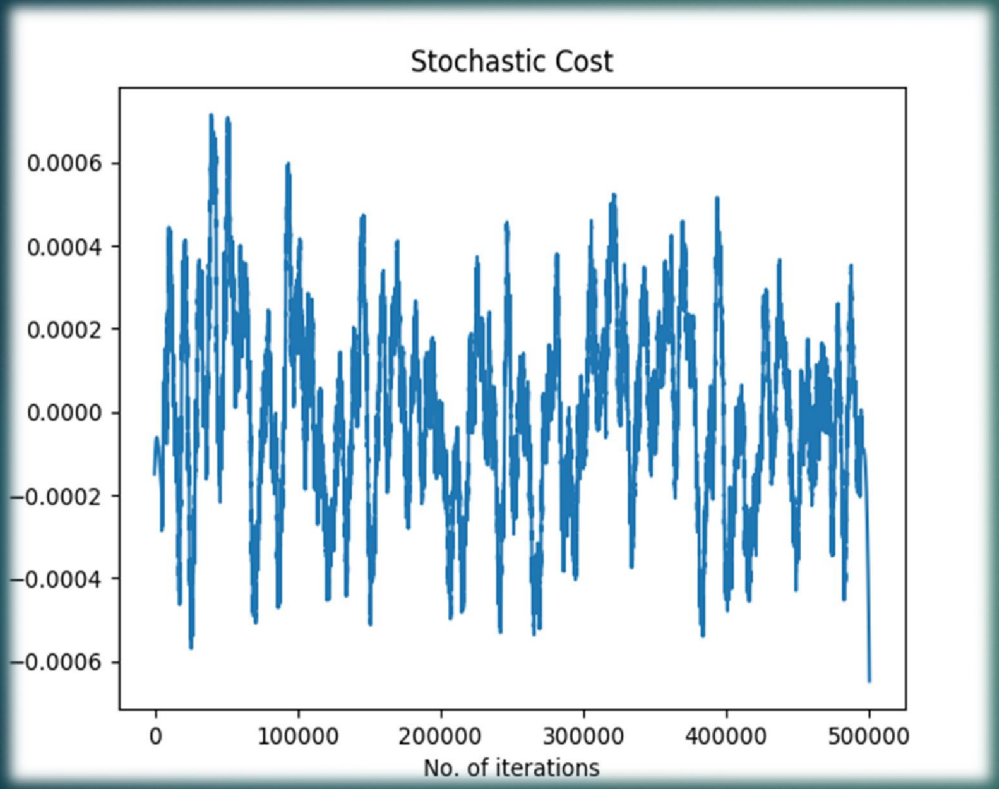
LEARNING RATE = .0001
ITERATIONS = 50,000

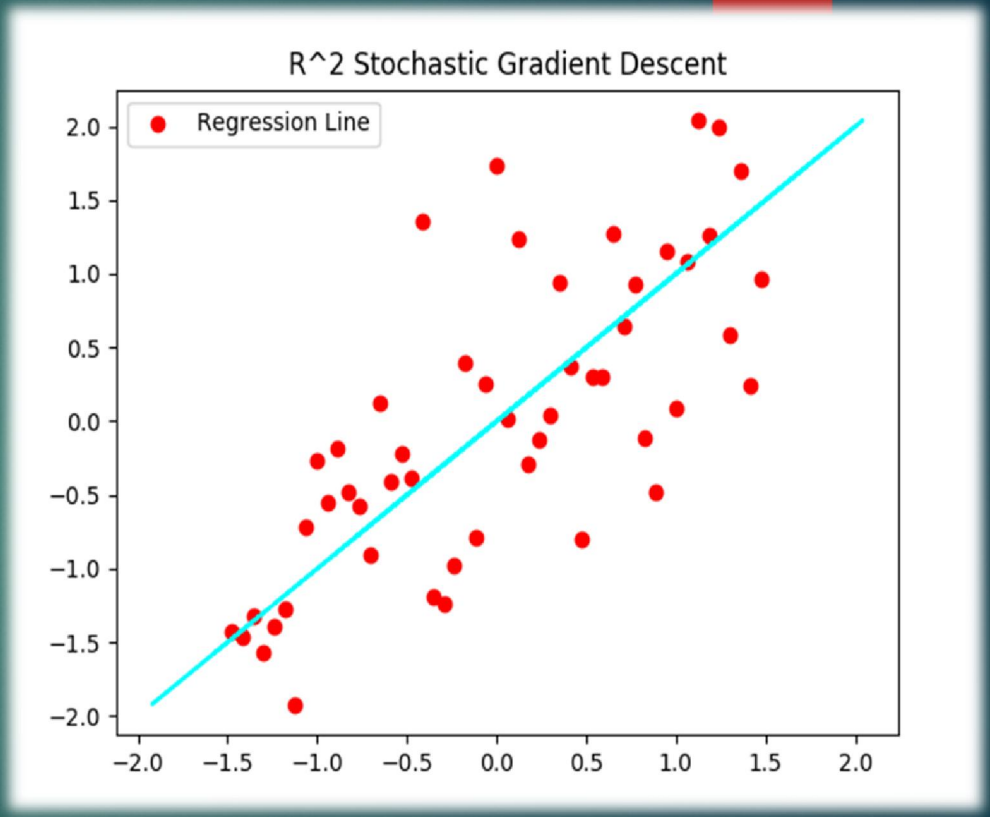
R^2 SCORE = .49
RESIDUAL SUM OF SQUARES = 2.02

STOCHASTIC GRADIENT DESCENT

- Same procedure as before except first randomly shuffle the data set
- In this case each iteration calculates the gradient based on a randomly chosen sample, so batch size = 1.
- Remove noise from the cost curve

Stochastic Gradient Descent





LEARNING RATE = .0001
ITERATIONS = 500,000

R^2 SCORE = .39
RESIDUAL SUM OF SQUARES = 1.75

CONCLUSION

- K-NN Classifier can predict what region you are in based on median income and educational attainment with a 64 % accuracy.
- A linear regression model was used to predict median household income level based on educational attainment with a R^2 score of .49
- Gradient Descent's cost curve converges quickly while Stochastic Gradient Descent's cost curve fluctuates.
- Gradient Descent and K-NN are computationally expensive for larger data sets.

REFERENCES

- 1) HTTPS://WWW.ERS.USDA.GOV/DATA-PRODUCTS National Education Data up to 2016
- 2) HTTPS://www2.census.gov/programs-surveys/cps/tables Median Household Income by State
- 3) https://github.com/Columbia-Intro-Data-Science